

Decision-Relevant Modelling of Extreme Demand for Industrial Electricity Tariffs

Short title: Extreme Demand and Industrial Electricity Tariff Decisions

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ABSTRACT

Peak-sensitive electricity tariffs turn uncertainty in monthly maximum demand into a mixed decision over tariff mode and, under contract demand, the contracted level. Because competing modes depend on different functionals of the predictive distribution, accuracy at a single upper quantile need not determine economic performance. We develop a tariff-specific stochastic framework in which contract-demand optimization is a constrained Expected-Shortfall problem, full-action regret separates mode-selection loss from residual within-contract loss, and direct tariff contrasts yield tighter Wasserstein stability margins by retaining the common distributional perturbation of competing costs. In 10,000 paired 24-month histories with five predictive procedures and nine tariff settings (450,000 decisions), two methods fitted to the same monthly maxima reverse their economic ranking: the GEV-minus-KDE mean-regret contrast is -3.887 thousand CNY/month in the contract-demand region and $+5.470$ thousand CNY/month in the capacity region. Alternative count-severity and physical-pipeline mechanisms also change relative performance. Rolling forecasts for eight manufacturing facilities produce score-dependent rankings with substantial factory-level uncertainty. The economic value of a peak-demand forecast is therefore governed jointly by the functional exposed by the tariff, the separation of competing costs, and the action induced by the predictive distribution.

KEYWORDS

Extreme value theory; probabilistic forecasting; industrial electricity demand; demand charges; stochastic optimization

1 | INTRODUCTION

Electrification is increasing the operational importance of peak-sensitive electricity pricing for industrial energy management. Capacity subscriptions, contracted-demand penalties, and other peak-sensitive tariffs require users to commit to a pricing option or contracted level before future peak load is known, so uncertainty in demand can alter both the preferred action and its expected cost¹⁻⁴. Peak-demand forecasting is therefore an economic decision problem as well as a statistical prediction problem. For an industrial user, the relevant question is which features of the predictive distribution govern the tariff action and how errors in those features propagate into cost.

Research relevant to this question has developed along three complementary lines. Contracting and capacity-subscription studies optimize subscribed or contracted demand under uncertain load and examine the sensitivity of contract choices to demand and price errors^{1-3,5,6}. Probabilistic-forecasting research places predictive distributions inside downstream operational problems, including reserve planning, market bidding and scheduling, and decision-focused forecast combination⁷⁻⁹. A related application-driven literature evaluates forecasts through their operational or economic consequences and shows that statistical and decision-based rankings can differ^{10,11}. This literature establishes that predictive information and downstream cost must be studied jointly when forecasts are used for operational decisions.

The remaining structural question for industrial tariffs concerns how an error in one predictive distribution propagates through several competing charging modes. Capacity charging has a fixed mode value, actual-demand charging depends on the predictive mean, and contract-demand charging combines an exceedance functional with optimization over the contract level. The complete tariff action therefore couples a continuous decision within contract demand with a discrete comparison of internally optimized mode values. Existing demand-contract analyses characterize contract levels and their sensitivity to forecast inputs, while decision-focused studies evaluate the value of forecasts for a specified downstream problem^{6,8,9}. The remaining link is a distribution-to-action account that preserves the shared perturbation across competing tariff costs, distinguishes mode switching from residual contract-level loss, and expresses the resulting stability on the physical demand scale.

We develop such an account for a three-mode industrial tariff. The contract-demand optimization is identified with a constrained Expected-Shortfall functional at a tariff-implied probability level, and full-action regret is decomposed exactly into mode-selection loss and within-contract excess. Direct analysis of competing tariff costs yields contrast-specific Wasserstein stability constants; for actual-demand versus contract-demand charging, the shared predictive law gives a tighter bound than treating the two optimized mode values separately. A controlled study with 10,000 paired histories, five predictive constructions, and nine primary tariff cells evaluates these mechanisms through information-matched comparisons and targeted stress designs. Rolling forecasts from eight manufacturing facilities provide an external evaluation of predictive scores and hypothetical tariff charges. The framework links distributional estimation, tariff geometry, and downstream economic loss within a single stochastic decision problem.

2 | TARIFF AND DECISION FRAMEWORK

2.1 | Tariff-relevant distributional functionals

The customer makes a one-period, risk-neutral tariff decision before monthly maximum demand is observed. The reference capacity S is fixed within a tariff cell and serves as a billing parameter. Under contract-demand charging, the customer pays $r_d D$ plus $r_d \kappa (M - \alpha D)_+$, so αD is the threshold for the additional demand charge. The expected costs below compare these demand-charge components under a common predictive law.

Let $M \geq 0$ denote monthly maximum demand and let F be its true distribution. Suppose $\mathbb{E}_F[M] < \infty$. Define the mean functional

$$\mu_F = \mathbb{E}_F[M] \quad (1)$$

and the stop-loss transform

$$H_F(t) = \mathbb{E}_F[(M - t)_+] = \int_t^\infty \{1 - F(x)\} dx, \quad t \geq 0. \quad (2)$$

$H_F(t)$ is the expected exceedance above t and integrates the entire survival curve beyond that threshold.

For fixed tariff parameters and reference capacity S , consider the actions a_{cap} , a_{act} , and $a_{\text{con}}(D)$, with $D \in [\delta S, S]$ under the contract-demand mode. The demand quantities M , D , and S are defined on a common physical basis; in the controlled experiment below, this basis is kW. Their expected monthly costs are

$$C_F(a_{\text{cap}}) = r_c S,$$

$$C_F(a_{\text{act}}) = r_d \mu_F,$$

and

$$C_F(a_{\text{con}}(D)) = r_d \{D + \kappa H_F(\alpha D)\}.$$

The functionals in Equations (1)–(2) determine the expected costs. The corresponding tariff modes, decision variables, expected costs, and distributional functionals are summarized in Table 1.

For finite-mean F , Table 1 shows that the capacity-mode cost is independent of F , actual-demand cost depends on μ_F , and contract-demand cost at a given D depends on $H_F(\alpha D)$. Optimizing the contract level and comparing its cost with other modes requires the stop-loss curve over the feasible thresholds. We use $q_{0.99}$ error as a complementary measure of upper-tail fidelity.

2.2 | Contract-demand optimization

Contract-demand charging couples the choice of contract level to the relevant distributional functional: D determines the threshold αD at which the stop-loss transform is evaluated. Let the feasible contract set be

$$\mathcal{D}_S = [\delta S, S].$$

For $D \in \mathcal{D}_S$, the expected monthly contract-demand cost is

$$C_F(a_{\text{con}}(D)) = r_d \{D + \kappa H_F(\alpha D)\}.$$

70 The set of optimal contract levels and the corresponding optimized mode value are

$$71 \quad \mathcal{D}_F^* = \arg \min_{D \in \mathcal{D}_S} C_F(a_{\text{con}}(D))$$

72 and

$$73 \quad V_F(\text{con}) = \min_{D \in \mathcal{D}_S} r_d \{D + \kappa H_F(\alpha D)\}.$$

74 **Proposition 1** (Contract-demand optimization). *Suppose $M \geq 0$, $\mathbb{E}_F[M] < \infty$, $S > 0$, $r_d > 0$, $\alpha > 0$, $\kappa > 0$,*
 75 *and $0 \leq \delta \leq 1$. Let*

$$76 \quad \phi_F(D) = D + \kappa H_F(\alpha D), \quad D \in \mathcal{D}_S. \quad (3)$$

77 *Then ϕ_F is finite, continuous, and convex on $\mathcal{D}_S$. Since $\mathcal{D}_S$ is compact,*

$$78 \quad \arg \min_{D \in \mathcal{D}_S} C_F(a_{\text{con}}(D)) = \arg \min_{D \in \mathcal{D}_S} \phi_F(D)$$

79 *is nonempty. For any $D \in (\delta S, S)$,*

$$80 \quad \partial \phi_F(D) = \left[1 - \kappa \alpha \Pr_F(M \geq \alpha D), 1 - \kappa \alpha \Pr_F(M > \alpha D) \right].$$

81 *Hence every interior optimizer $D^* \in (\delta S, S)$ satisfies*

$$82 \quad \Pr_F(M > \alpha D^*) \leq \frac{1}{\kappa \alpha} \leq \Pr_F(M \geq \alpha D^*).$$

83 *When $\kappa \alpha > 1$, this is equivalently*

$$84 \quad F(\alpha D^* -) \leq 1 - \frac{1}{\kappa \alpha} \leq F(\alpha D^*). \quad (4)$$

85 *If, in addition, F is continuous at αD^* , then*

$$86 \quad F(\alpha D^*) = 1 - \frac{1}{\kappa \alpha}.$$

87 *If $\kappa \alpha > 1$, define*

$$88 \quad p^* = 1 - \frac{1}{\kappa \alpha}, \quad I_S = [\alpha \delta S, \alpha S].$$

89 *For $p \in (0, 1)$ and a nonempty closed interval I , define the constrained Expected Shortfall functional*

$$90 \quad \text{ES}_{p,I}(F) = \min_{t \in I} \left\{ t + \frac{1}{1-p} \mathbb{E}_F[(M-t)_+] \right\}.$$

91 *Then the optimized contract-demand value satisfies*

$$92 \quad V_F(\text{con}) = \frac{r_d}{\alpha} \text{ES}_{p^*, I_S}(F). \quad (5)$$

93 *Let*

$$94 \quad Q_p(F) = \{t \geq 0 : F(t-) \leq p \leq F(t)\}$$

95 *be the generalized p -quantile set, and define ordinary Expected Shortfall by*

$$96 \quad \text{ES}_p(F) = \frac{1}{1-p} \int_p^1 F^{-1}(u) du, \quad F^{-1}(u) = \inf\{x : F(x) \geq u\}.$$

97 If $Q_{p^*}(F) \cap I_S \neq \emptyset$, then

$$98 \quad V_F(\text{con}) = \frac{r_d}{\alpha} \text{ES}_{p^*}(F), \quad (6)$$

99 and the feasible optimal thresholds are exactly $Q_{p^*}(F) \cap I_S$.

100 The subgradient expression follows from the one-sided derivatives of the stop-loss transform,

$$101 \quad H'_{F,-}(t) = -\Pr_F(M \geq t), \quad H'_{F,+}(t) = -\Pr_F(M > t).$$

102 The convex objective in Equation (3) therefore gives the quantile condition in Equation (4).

103 When $\delta < 1$, the corresponding boundary conditions follow from the one-sided derivatives of the convex
104 objective. The lower boundary $D^* = \delta S$ is optimal if and only if

$$105 \quad 1 - \kappa\alpha \Pr_F(M > \alpha\delta S) \geq 0,$$

106 whereas the upper boundary $D^* = S$ is optimal if and only if

$$107 \quad 1 - \kappa\alpha \Pr_F(M \geq \alpha S) \leq 0.$$

108 If $\kappa\alpha < 1$, ϕ_F is strictly increasing on any nondegenerate feasible interval, so the lower boundary is the
109 unique optimizer. If $\kappa\alpha = 1$, ϕ_F is nondecreasing; the lower boundary is therefore optimal, although a
110 flat region may yield additional minimizers. These cases are retained by the general argmin formulation
111 above.

112 For the Expected-Shortfall representation, set $t = \alpha D$. Since $1/(1 - p^*) = \kappa\alpha$,

$$113 \quad D + \kappa \mathbb{E}_F[(M - \alpha D)_+] = \frac{1}{\alpha} \left\{ t + \frac{1}{1 - p^*} \mathbb{E}_F[(M - t)_+] \right\}.$$

114 Minimizing over $D \in [\delta S, S]$ gives Equation (5). The standard Rockafellar–Uryasev variational represen-
115 tation of Expected Shortfall¹² gives Equation (6) when the generalized quantile set intersects the feasible
116 threshold interval. Here both the probability level p^* and the feasible threshold interval I_S are fixed by
117 the tariff coefficients. The representation therefore separates two roles of the upper tail: a generalized p^* -
118 quantile locates an optimal contract threshold, whereas the corresponding upper-tail average determines
119 the optimized contract-mode value whenever the quantile-optimal set is feasible.

120 For the controlled tariff coefficients used below, $\kappa = 2$ and $\alpha \in \{1.05, 1.20\}$, giving

$$121 \quad p^* = 1 - \frac{1}{2(1.05)} \approx 0.524$$

122 and

$$123 \quad p^* = 1 - \frac{1}{2(1.20)} \approx 0.583,$$

124 respectively. For an interior optimum, $t_F^* = \alpha D_F^*$ satisfies the generalized p^* -quantile condition in
125 Equation (4); if F is continuous at t_F^* , then $F(t_F^*) = p^*$. If the unconstrained quantile-optimal threshold set
126 does not intersect the feasible interval $[\alpha\delta S, \alpha S]$, the constrained optimizer is instead determined by the
127 corresponding boundary condition above. Together with the selected contract level, the stop-loss value at
128 the selected threshold determines the optimized contract cost.

129 The connection to forecast scoring is exact within the contract mode. For $\kappa\alpha > 1$, define the pinball loss
130 at the tariff-implied level p^* by $\ell_{p^*}(z) = z\{p^* - \mathbf{1}(z < 0)\}$. For any demand realization y and feasible D ,

$$D + \kappa(y - \alpha D)_+ = \kappa \ell_{p^*}(y - \alpha D) + y/\alpha. \quad (7)$$

Equation (7) follows by expanding the pinball loss. Its final term is independent of D , so minimizing expected contract cost is equivalent to minimizing expected pinball loss over the feasible thresholds.

Within-contract loss. For any feasible $D \in \mathcal{D}_S$ and any $D_F^* \in \mathcal{D}_F^*$,

$$\phi_F(D) - \phi_F(D_F^*) = \int_{D_F^*}^D [1 - \kappa\alpha\{1 - F(\alpha u)\}] du.$$

When $\kappa\alpha > 1$, let

$$t = \alpha D, \quad t_F^* = \alpha D_F^*.$$

Then

$$C_F(a_{\text{con}}(D)) - C_F(a_{\text{con}}(D_F^*)) = r_d \kappa \int_{t_F^*}^t \{F(s) - p^*\} ds. \quad (8)$$

These identities hold for interior or boundary optimizers and do not require continuity of F . Atoms create distinct left and right derivatives at the corresponding thresholds but do not alter the integral representation. If D_F^* is interior and F has a positive continuous density in a neighbourhood of t_F^* , the within-contract excess is locally quadratic in $t - t_F^*$. This local quadratic behaviour need not hold at boundary solutions or atoms.

Comparing this optimized contract with the other modes also requires its cost level and the mean demand.

The theoretical problem is posed on the continuous feasible set $\mathcal{D}_S$. Once the contract-demand mode and the two remaining tariff modes have been internally evaluated, the next question is whether their optimized values are sufficiently separated for the selected tariff mode to remain stable under distributional estimation error.

2.3 | Oracle mode separation and decision stability

After each tariff mode has been evaluated on its own terms, the tariff decision reduces to comparing a finite set of internally optimized mode values. Let

$$\mathcal{M} = \{\text{cap}, \text{act}, \text{con}\},$$

with

$$V_F(\text{cap}) = C_F(a_{\text{cap}}), \quad V_F(\text{act}) = C_F(a_{\text{act}}),$$

and $V_F(\text{con})$ defined by the contract-demand optimization in Section 2.2. The oracle tariff value is

$$V_F^* = \min_{m \in \mathcal{M}} V_F(m).$$

When the minimizing mode is unique, denote it by m_F^* . If $V_F^{(2)}$ is the second-lowest of the three optimized mode values, define

$$\gamma_F = V_F^{(2)} - V_F^* > 0. \quad (9)$$

Thus, γ_F measures separation between optimized tariff modes under the true distribution. It is a mode-level quantity, not a second-best gap over the continuous contract-level action space.

164 For an estimated distribution $\hat{F}$, define $V_{\hat{F}}(m)$ by applying the same tariff-cost and within-mode
 165 optimization map with F replaced by $\hat{F}$. Throughout this comparison, the tariff cell

$$166 \quad (S, r_c, r_d, \alpha, \kappa, \delta)$$

167 and hence the feasible action set are held fixed. Let

$$168 \quad \hat{m} \in \arg \min_{m \in \mathcal{M}} V_{\hat{F}}(m).$$

169 For each $m \in \mathcal{M}$, let $\mathcal{A}_m$ denote the feasible action set for that tariff mode, with

$$170 \quad \mathcal{A}_{\text{cap}} = \{a_{\text{cap}}\}, \quad \mathcal{A}_{\text{act}} = \{a_{\text{act}}\}, \quad \mathcal{A}_{\text{con}} = \{a_{\text{con}}(D) : D \in \mathcal{D}_S\},$$

171 and define the primitive action space

$$172 \quad \mathcal{A}_S = \bigcup_{m \in \mathcal{M}} \mathcal{A}_m.$$

173 Let $\hat{a} \in \mathcal{A}_{\hat{m}}$ be a full action associated with the selected mode $\hat{m}$ and optimal under $\hat{F}$, so that

$$174 \quad \hat{a} \in \arg \min_{a \in \mathcal{A}_S} C_{\hat{F}}(a).$$

175 When $\hat{m} = \text{con}$, write $\hat{a} = a_{\text{con}}(\hat{D})$. Define its full-action regret under the true distribution by

$$176 \quad R_F(\hat{a}) = C_F(\hat{a}) - V_F^*.$$

177 Then

$$178 \quad R_F(\hat{a}) = \underbrace{V_F(\hat{m}) - V_F^*}_{\text{mode-selection loss}} + \underbrace{C_F(\hat{a}) - V_F(\hat{m})}_{\text{within-mode excess}}. \quad (10)$$

179 Equation (10) is obtained by adding and subtracting $V_F(\hat{m})$ and does not require a unique oracle mode, an
 180 interior optimizer, or continuity of F . For the capacity and actual-demand modes, the feasible action sets
 181 are singletons, so the within-mode excess is zero. Only contract demand can have nonzero within-mode
 182 excess. In particular, if contract demand is both selected and oracle-optimal, the mode-selection term is
 183 zero and the remaining full-action regret is the within-contract loss in Equation (8), evaluated at $D = \hat{D}$.

184 The oracle quantities $V_F(m)$, V_F^* , and, when defined, m_F^* and γ_F are properties of the true distribution
 185 and do not enter construction of the estimated action.

186 To measure the largest distribution-induced cost perturbation before mode comparison, define

$$187 \quad \varepsilon(F, \hat{F}) = \sup_{a \in \mathcal{A}_S} |C_{\hat{F}}(a) - C_F(a)|.$$

188 Suppose that both F and $\hat{F}$ have finite first moments and that the tariff cell is held fixed. Because the
 189 capacity-mode cost is distribution-independent, while the common D term cancels when contract costs
 190 are compared at the same primitive action, the perturbation specializes to

$$191 \quad \varepsilon(F, \hat{F}) = r_d \max \left\{ |\mu_{\hat{F}} - \mu_F|, \kappa \sup_{D \in \mathcal{D}_S} |H_{\hat{F}}(\alpha D) - H_F(\alpha D)| \right\}, \quad (11)$$

192 for $r_d > 0$.

193 **Proposition 2** (Oracle mode preservation). *Suppose F and $\widehat{F}$ have finite first moments, all relevant tariff*
 194 *costs are finite, $\varepsilon(F, \widehat{F}) < \infty$, and the oracle tariff mode m_F^* is unique. For each $m \neq m_F^*$, define the pairwise*
 195 *oracle gap*

$$196 \quad \gamma_{F,m} = V_F(m) - V_F(m_F^*) > 0$$

197 *and, for each $m \in \mathcal{M}$, the optimized mode-value error*

$$198 \quad e_m = |V_{\widehat{F}}(m) - V_F(m)|.$$

199 *If*

$$200 \quad e_m + e_{m_F^*} < \gamma_{F,m} \quad \text{for every } m \neq m_F^*,$$

201 *then*

$$202 \quad \arg \min_{m \in \mathcal{M}} V_{\widehat{F}}(m) = \{m_F^*\}.$$

203 *In particular, the uniform condition*

$$204 \quad \gamma_F > 2\varepsilon(F, \widehat{F})$$

205 *is sufficient for the same conclusion.*

206 *For every competitor $m \neq m_F^*$,*

$$207 \quad \begin{aligned} V_{\widehat{F}}(m) - V_{\widehat{F}}(m_F^*) &\geq V_F(m) - e_m - \{V_F(m_F^*) + e_{m_F^*}\} \\ &= \gamma_{F,m} - e_m - e_{m_F^*}. \end{aligned}$$

208 *The pairwise condition therefore makes every estimated competing mode strictly more costly than m_F^* .*

209 *Moreover, uniform perturbation of the primitive-action costs implies*

$$210 \quad e_m \leq \varepsilon(F, \widehat{F}) \quad \text{for every } m \in \mathcal{M}.$$

211 *For contract demand, this follows from the standard bound for minima of uniformly perturbed objectives*
 212 *and does not require the optimizing contract levels under F and $\widehat{F}$ to coincide. Since, for a unique oracle*
 213 *mode,*

$$214 \quad \gamma_F = \min_{m \neq m_F^*} \gamma_{F,m},$$

215 *the condition $\gamma_F > 2\varepsilon(F, \widehat{F})$ implies all of the pairwise inequalities above. Both conditions are sufficient*
 216 *rather than necessary.*

217 *Remark.* When the oracle minimum is not unique, Proposition 2 is interpreted set-wise: stability excludes
 218 *positive-gap nonoptimal modes rather than preserving an arbitrary label among tied oracle minimizers.*

219 *For a uniform mode-value summary, define*

$$220 \quad e_V = \max_{m \in \mathcal{M}} e_m = \max_{m \in \mathcal{M}} |V_{\widehat{F}}(m) - V_F(m)|.$$

221 *The preceding bound gives*

$$222 \quad e_V \leq \varepsilon(F, \widehat{F}).$$

223 **Corollary 1** (Mode-value error certificate). *Suppose the oracle tariff mode m_F^* is unique and define*

$$224 \quad d_{MV} = \frac{2e_V}{\gamma_F}.$$

225 *If*

$$226 \quad d_{MV} < 1,$$

227 *then*

$$228 \quad \arg \min_{m \in \mathcal{M}} V_{\hat{F}}(m) = \{m_F^*\}.$$

229 *Consequently, any tariff-mode error must satisfy*

$$230 \quad d_{MV} \geq 1.$$

231 Indeed, $d_{MV} < 1$ gives

$$232 \quad e_m + e_{m_F^*} \leq 2e_V < \gamma_F \leq \gamma_{F,m}$$

233 for every $m \neq m_F^*$, so Proposition 2 applies. The certificate replaces the pair-specific errors and gaps by the
 234 uniform quantities e_V and γ_F , making it simpler but more conservative than the pairwise condition. Both
 235 e_V and γ_F depend on oracle quantities under F . Accordingly, d_{MV} is used in the controlled experiment
 236 as an oracle-dependent retrospective mode-stability certificate, not as an independent diagnostic directly
 237 available to an industrial decision maker at deployment.

238 Mode preservation is distinct from preservation of the full tariff action. For the exact $\hat{F}$ -optimal full
 239 action $\hat{a}$ defined above, optimality under $\hat{F}$ and the uniform primitive-action perturbation give

$$240 \quad 0 \leq R_F(\hat{a}) \leq 2\varepsilon(F, \hat{F}). \quad (12)$$

241 The exact decomposition in Equation (10) sharpens this statement when the selected tariff mode is
 242 oracle optimal. If the selected oracle-optimal mode is actual demand or capacity, the within-mode action
 243 is fixed and hence

$$244 \quad R_F(\hat{a}) = 0.$$

245 Only an oracle-optimal contract-demand mode can retain positive within-mode excess. Define

$$246 \quad \Delta_H(F, \hat{F}) = \sup_{D \in \mathcal{D}_S} |H_{\hat{F}}(\alpha D) - H_F(\alpha D)|.$$

247 If contract demand is both selected and oracle optimal, and $\hat{D}$ exactly minimizes the contract objective
 248 under $\hat{F}$, then

$$249 \quad 0 \leq R_F(a_{\text{con}}(\hat{D})) \leq 2r_d \kappa \Delta_H(F, \hat{F}). \quad (13)$$

250 Thus, once an oracle-optimal contract mode is selected, its residual full-action loss is controlled by pertur-
 251 bation of the stop-loss curve over the feasible contract thresholds. Equation (8) gives the corresponding
 252 exact loss representation for the selected contract level.

253 For a numerical action $\tilde{a}$ that is not required to attain the fitted optimum exactly, suppose that, for some
 254 $\zeta \geq 0$,

$$255 \quad C_{\hat{F}}(\tilde{a}) \leq \inf_{a \in \mathcal{A}_S} C_{\hat{F}}(a) + \zeta.$$

256 Then

$$257 \quad 0 \leq R_F(\tilde{a}) \leq 2\varepsilon(F, \hat{F}) + \zeta. \quad (14)$$

258 Here ζ is an objective-suboptimality allowance in the same monthly tariff-cost units as C_F ; its numerical
 259 construction and implementation audit are reported in the Supporting Information.

2.4 | Tariff-contrast Wasserstein stability

For two nonnegative distributions with finite first moments, define their one-dimensional Wasserstein distance by

$$W_1(F, G) = \int_0^\infty |F(x) - G(x)| dx. \quad (15)$$

The mean and stop-loss identities imply

$$|\mu_G - \mu_F| \leq W_1(F, G), \quad |H_G(t) - H_F(t)| \leq W_1(F, G) \quad \text{for every } t \geq 0.$$

For a fixed tariff cell, the optimized mode values therefore satisfy

$$|V_G(m) - V_F(m)| \leq L_m W_1(F, G), \quad L_{\text{cap}} = 0, \quad L_{\text{act}} = r_d, \quad L_{\text{con}} = r_d \kappa. \quad (16)$$

The capacity value is distribution-independent. For actual demand, the bound follows from the mean inequality. For contract demand, the two objectives differ uniformly over the common feasible set $\mathcal{D}_S$ by at most $r_d \kappa W_1(F, G)$, so the same bound holds for their minima without requiring the optimal contract levels under F and G to coincide.

The mode-wise bound can be sharpened by comparing tariff contrasts directly. For distinct modes $m, n \in \mathcal{M}$, define

$$\Delta_{F,mn} = V_F(m) - V_F(n).$$

The direct contrast constants are

$$K_{\text{cap,act}} = r_d, \quad K_{\text{cap,con}} = r_d \kappa, \quad K_{\text{act,con}} = r_d \max\{1, |1 - \kappa|\}, \quad (17)$$

with $K_{mn} = K_{nm}$. These constants yield the following result.

Proposition 3 (Tariff-contrast Wasserstein stability). *Suppose F and G are nonnegative distributions with finite first moments, the tariff cell is fixed, and the tariff coefficients and feasible contract set are common to both distributions. Then, for every distinct pair $m, n \in \mathcal{M}$,*

$$|\Delta_{G,mn} - \Delta_{F,mn}| \leq K_{mn} W_1(F, G). \quad (18)$$

If the oracle mode m_F^* under F is unique, define

$$\rho_F^{\text{contrast}} = \min_{m \neq m_F^*} \frac{\gamma_{F,m}}{K_{m,m_F^*}}. \quad (19)$$

Then

$$W_1(F, G) < \rho_F^{\text{contrast}} \implies \arg \min_{m \in \mathcal{M}} V_G(m) = \{m_F^*\}. \quad (20)$$

Moreover,

$$\rho_F^{\text{contrast}} \geq \rho_F^{\text{pair}} \geq \rho_F,$$

where ρ_F^{pair} and ρ_F are the mode-wise pairwise and uniform radii defined below.

For capacity–actual and capacity–contract comparisons, Equation (18) follows from the mean and contract-value bounds, respectively. For actual versus contract demand, let

$$g_D(x) = x - D - \kappa(x - \alpha D)_+.$$

292 Then

$$293 \quad V_F(\text{act}) - V_F(\text{con}) = r_d \sup_{D \in \mathcal{D}_S} \mathbb{E}_F[g_D(M)].$$

294 The two slopes of g_D are 1 and $1 - \kappa$, so its Lipschitz constant is $\max\{1, |1 - \kappa|\}$, uniformly in D . The
 295 Kantorovich–Rubinstein inequality gives

$$296 \quad |\mathbb{E}_G[g_D(M)] - \mathbb{E}_F[g_D(M)]| \leq \max\{1, |1 - \kappa|\} W_1(F, G).$$

297 Taking suprema over the same feasible contract set preserves this uniform bound and proves the actual–
 298 contract case of Equation (18). This argument does not require the contract optimizers under F and G to
 299 coincide.

300 The tightening is specific to contrasts whose two mode values respond to the predictive law. Capacity is
 301 distribution-independent, so the direct capacity–actual and capacity–contract constants coincide with their
 302 separate-mode counterparts. Actual demand and contract demand share the same distributional perturba-
 303 tion, and direct comparison retains cancellation that is lost when their mode values are bounded separately.
 304 With $\kappa = 2$, the actual–contract constant is r_d , compared with $3r_d$ under the separate mode-value bound.

305 Suppose the oracle mode m_F^* is unique. Combining Equation (16) with Proposition 2, the mode-wise
 306 pairwise conditions

$$307 \quad (L_m + L_{m_F^*})W_1(F, G) < \gamma_{F,m} \quad \text{for every } m \neq m_F^*$$

308 are sufficient for preserving the oracle mode. Define the corresponding demand-scale radius by

$$309 \quad \rho_F^{\text{pair}} = \min_{m \neq m_F^*} \frac{\gamma_{F,m}}{L_m + L_{m_F^*}}. \quad (21)$$

310 Then

$$311 \quad W_1(F, \hat{F}) < \rho_F^{\text{pair}} \implies \hat{m} = m_F^*. \quad (22)$$

312 This is a sufficient, not necessary, mode-preservation radius. Both W_1 and the radii are expressed in
 313 demand units, which are kW in the controlled experiment.

314 For a simpler uniform representation, let

$$315 \quad L = r_d \max\{1, \kappa\}.$$

316 Then

$$317 \quad \varepsilon(F, G) \leq LW_1(F, G). \quad (23)$$

318 Since $L_m + L_{m_F^*} \leq 2L$ and $\gamma_{F,m} \geq \gamma_F$, the uniform radius

$$319 \quad \rho_F = \frac{\gamma_F}{2L} \quad (24)$$

320 is a simpler conservative sufficient radius. For an exact $\hat{F}$ -optimal action $\hat{a}$, the same uniform bound gives

$$321 \quad R_F(\hat{a}) \leq 2\varepsilon(F, \hat{F}) \leq 2LW_1(F, \hat{F}).$$

322 The contrast-aware radius controls sufficient preservation of the tariff mode, whereas the uniform
 323 perturbation bound also supplies a general control on full-action regret.

324 The physical-clipping robustness analysis uses corresponding mean and stop-loss identities. Their
 325 derivation and the relation between event- and monthly-maximum clipping are given in Appendix S1.9.

3 | DISTRIBUTION ESTIMATION AND TARIFF ACTIONS

3.1 | Distribution estimators

Given an estimated distribution $\hat{F}$, the tariff map in Section 2 requires the predictive functionals associated with the competing tariff modes. We compare five predictive constructions organized by their information source. TAIL and EVENT-EMP use the event-level history, whereas GEV, KDE, and EMP use the 24 observed monthly maxima. This design compares complete forecasting procedures and also compares methods that use the same information. TAIL and EVENT-EMP represent alternative models of the event history; GEV, KDE, and EMP provide parametric, smooth nonparametric, and empirical models of monthly maxima.

TAIL uses empirical event-level threshold quantiles

$$0.90, \quad 0.925, \quad 0.95, \quad 0.975.$$

For a 24-month history, the first 19 months and final five months are used to select a threshold. Candidate thresholds are compared using held-out predictive performance together with prespecified goodness-of-fit, support, and shape-stability gates. Let Z denote an event-level maximum and u a candidate threshold. Strict exceedances,

$$Y = Z - u \mid Z > u,$$

are modelled by a generalized Pareto distribution, with shape and scale estimated by maximum likelihood and location fixed at zero. Monthly exceedance counts are modelled by a Poisson distribution when empirical dispersion is limited and by a negative-binomial distribution otherwise. In the controlled experiment, event-level maxima are treated as distinct events, so no additional clustering aggregation is applied at this stage.

After threshold selection, the numerical threshold is fixed and the GPD, count law, and subthreshold monthly component are refitted using the complete 24-month history. The final fit must satisfy $\hat{\xi} < 1$, ensuring that the GPD component and the tariff-relevant mean and stop-loss functionals are finite. The additional condition $\hat{\xi} < 0.5$ is recorded as a finite-variance stability flag. Candidate fits are screened for exceedance support, shape uncertainty, tail goodness of fit, and compatibility of any fitted finite GPD endpoint with the observed event support. When the primary GPD specification does not satisfy the prespecified finite-mean or support criteria, the fallback hierarchy uses an exponential tail when sufficient exceedance information remains and otherwise uses the empirical monthly-maxima distribution. The primary screening requires at least 100 exceedances, a 95% shape-interval width at most 0.8, and a uniform-transform goodness-of-fit p -value of at least 0.01, together with the finite-mean and support checks. When no eligible primary candidate remains, a threshold with at least 50 exceedances supports an exponential-tail fit; otherwise the procedure uses the empirical monthly-maxima law. Appendix S1.3 specifies the selection score, refitting checks, and deterministic tie rules.

For the final TAIL law, the empirical distribution of monthly subthreshold maxima is combined with the fitted exceedance-count distribution and the GPD excess distribution to define the monthly-maximum law.

EVENT-EMP provides a nonparametric event-level construction from the same 24-month event history. Let $\hat{G}_E$ denote the pooled empirical distribution function of event-level maxima and let N_m denote the observed total number of events in month m . Conditional on N_m , the induced monthly-maximum distribution

365 has CDF $\widehat{G}_E(x)^{N_m}$. Averaging over the empirical distribution of monthly event counts gives

$$366 \quad \widehat{F}_{EE}(x) = \frac{1}{24} \sum_{m=1}^{24} \widehat{G}_E(x)^{N_m}.$$

367 EVENT-EMP therefore propagates the empirical event-value and monthly event-count distributions di-
 368 rectly to monthly maximum demand. Together with TAIL, it provides an event-level reference for
 369 examining how different distributional constructions translate the same underlying event history into
 370 tariff-relevant predictive laws.

371 The remaining three methods use the 24 observed monthly maxima. GEV fits a generalized extreme-
 372 value distribution using sample probability-weighted moments to construct the first three L-moments,
 373 from which the distribution parameters are obtained deterministically. The shape search is restricted to a
 374 prespecified interval contained within the finite-mean region. For numerical evaluation, the fitted law is
 375 conditioned on the nonnegative portion of its natural support. Fits failing the parameter or support checks
 376 revert to the empirical monthly-maxima distribution.

377 KDE is fitted to the 24 observed monthly maxima using a Gaussian kernel with bandwidth

$$378 \quad h = 1.06 s n^{-1/5},$$

379 where s is the sample standard deviation and n the number of monthly maxima. The equal-weight Gaussian
 380 mixture is conditioned on nonnegative support. A numerically degenerate training sample triggers an
 381 empirical fallback.

382 EMP assigns equal mass to the observed monthly maxima.

383 **3.2 | From estimated distributions to tariff actions and regret**

384 All five fitted predictive distributions enter the tariff-action map of Section 2 through deterministic eval-
 385 uation of their means, stop-loss transforms, and quantiles. EMP and EVENT-EMP use exact finite sums,
 386 KDE uses truncated-normal expressions, and GEV and TAIL use deterministic inversion and integration
 387 where required. Predictive simulation is confined to internal specification selection for TAIL and does not
 388 enter the final tariff functionals or regret calculations.

389 Each fitted law is then evaluated over the continuous feasible interval $\mathcal{D}_S = [\delta S, S]$ using the contract
 390 objective in Section 2.2. The convex objective is minimized from its boundary and generalized-quantile
 391 conditions, using exact adjacent support points for discrete laws and deterministic root finding for con-
 392 tinuous laws. We resolve exact tariff-mode ties and nonunique discrete quantiles using the prespecified
 393 deterministic conventions in Appendix S1.4.

394 The selected tariff mode is

$$395 \quad \widehat{m} \in \arg \min_{m \in \mathcal{M}} \widehat{V}(m).$$

396 If $\widehat{m} = \text{con}$, let $\widehat{D}$ denote the selected minimizing contract level and define

$$397 \quad \widehat{a} = a_{\text{con}}(\widehat{D}).$$

For $\hat{m} = \text{cap}$ or $\hat{m} = \text{act}$, the selected action is a_{cap} or a_{act} , respectively, and D has no economic role in that action. Oracle functionals and actions are evaluated from the analytic DGP distribution using deterministic root finding and quadrature. Thus a fixed simulated truth sample does not contribute Monte Carlo error to the oracle mode, gap, action, or regret.

Two decision endpoints are kept distinct. In cells with a unique oracle mode, tariff-mode correctness is the row-level indicator $I(\hat{m} = m_F^*)$, evaluating only the discrete mode choice. Oracle-mode accuracy denotes the percentage of correct mode selections across Monte Carlo replications within a scenario–method cell, reported as “Acc.” in the scenario tables. Economic performance is instead assessed through full-action regret under the true distribution,

$$R_F(\hat{a}) = C_F(\hat{a}) - V_F^*.$$

Under contract demand, this endpoint includes the cost of the selected contract level as well as the selected mode. Numerically, regret is recorded as

$$\max \{0, C_F(\hat{a}) - V_F^*\},$$

where truncation at zero only guards against negative floating-point differences.

The comparison varies the predictive distribution while holding the tariff decision rule fixed.

4 | EVALUATION DESIGN AND STATISTICAL INFERENCE

4.1 | Controlled tariff experiment

The controlled experiment examines how errors in an estimated distribution affect decisions across tariff settings. We vary the tariff parameters to span different oracle modes and degrees of separation between optimized costs. Reusing each fitted distribution across these settings isolates the role of the tariff parameters in translating a given forecast into an action. The design uses a stylized tariff family to study this relationship.

Monthly maximum demand is generated from a Poisson–GPD construction. Within each month, 200 bulk events are drawn as

$$45,000 + (80,000 - 45,000)B, \quad B \sim \text{Beta}(8, 2),$$

in kW. Independently, the number of tail events satisfies

$$N \sim \text{Poisson}(30),$$

and each tail event has demand

$$80,000 + Y, \quad Y \sim \text{GPD}(\xi = 0.15, \sigma = 12,000),$$

with location fixed at zero for the excess distribution. Monthly maximum demand M is the maximum over all events generated in that month. No additional seasonal or amplitude component is included in the baseline data-generating process.

Each Monte Carlo replication contains a 24-month training history shared by TAIL, EVENT-EMP, GEV, KDE, and EMP. Each method produces one estimated distribution from that history, and the same method-specific estimate is reused across all tariff cells in the replication. The primary experiment contains 10,000 independent replications, producing 10,000 fitted distributions per method and preserving replication-level pairing across methods. Screening and fallback rules are stated in Appendix S1.

The controlled tariff coefficients are

$$r_d = 36, \quad r_c = 22.5, \quad \kappa = 2, \quad \delta = 0.40,$$

where r_d and r_c are expressed in CNY/(kW·month) and all demand quantities share the same kW basis. The reference capacity on this common demand basis is indexed by utilization,

$$u = \frac{\mu_F}{S}, \quad S = \frac{\mu_F}{u},$$

and therefore varies across tariff cells. The nine cells form two distinct design blocks:

$$\alpha = 1.05 : \quad u \in \{0.620, 0.630\},$$

and

$$\alpha = 1.20 : \quad u \in \{0.400, 0.450, 0.455, 0.625, 0.630, 0.635, 0.700\}.$$

The two blocks are not a continuous sweep over α . In particular, $\alpha = 1.20$ is used as a controlled stress setting for the decision mechanism and is not interpreted as a verified current regional tariff parameter.

The tariff cells were fixed from the oracle tariff geometry before method-specific regret outcomes were compared; estimator-specific performance did not enter cell selection. The $\alpha = 1.05$ cells place the actual-demand and capacity cases near mode boundaries at $u = 0.620$ and 0.630 , respectively. Under $\alpha = 1.20$, the selected cells span actual-demand ($u = 0.400, 0.450$), contract-demand ($u = 0.455, 0.625, 0.630$), and capacity ($u = 0.635, 0.700$) oracle regions with different degrees of optimized-mode separation. The cells cover the three oracle modes and a range of mode separations, including near-boundary and more clearly separated configurations. Here, “near” and “clear” refer only to relative separation among oracle optimized mode values. The design therefore exposes the same fitted distribution to different tariff-functional and mode-separation geometries without claiming causal identification of either mechanism in isolation. Regional contrasts summarize performance over these prespecified evaluated cells within each oracle region; they are not estimates of an average over the entire continuous oracle region.

Let F_B denote the CDF of one beta-bulk event and F_Z the CDF of one shifted-GPD tail event. Independence and the Poisson probability-generating function give the monthly-maximum oracle CDF

$$F_M(x) = F_B(x)^{200} \exp\{30[F_Z(x) - 1]\}.$$

Oracle quantiles are obtained by deterministic inversion, and μ_F and $H_F(t)$ by deterministic integration of $1 - F_M$. The analytic values are $\mu_F = 148,235.371489$ kW, $q_{0.99} = 265,663.010007$ kW, and $q_{0.999} = 375,513.529996$ kW. For the $\alpha = 1.20$ block, tariff geometry is evaluated deterministically over utilization; those evaluation locations are not additional forecasts or replications.

464 The five predictive constructions are evaluated over the crossed tariff grid

465
$$\alpha \in \{1.05, 1.10, 1.15, 1.20\}, \quad u \in \{0.40, 0.42, \dots, 0.76\},$$

466 which contains 76 tariff cells. Across 1,000 replications, the design produces

467
$$1,000 \times 5 \times 76 = 380,000$$

468 decision rows. A separate gap-calibrated analysis uses descriptive nearest-gap cell comparisons selected
469 from the oracle margins; these comparisons are not treated as one-to-one matched pairs or as independent
470 observations. Oracle contract-demand cells occur at $\alpha = 1.20$ only (nine cells); at each of the other three
471 α values, the oracle regions are actual-demand and capacity. The 380,000 rows are method–history–cell
472 decisions, not independent replications; the independent Monte Carlo units remain the 1,000 histories.

473 Section 4.2 next defines the statistical and decision endpoints and their corresponding inferential units.

474 4.2 | Evaluation endpoints and statistical inference

475 Performance is evaluated at several points in the distribution-to-decision chain, with the analysis unit
476 matched to the corresponding estimand. Statistical fidelity is summarized by absolute relative errors in
477 the mean and tariff-specific stop-loss functional, together with the 0.99-quantile error as a complementary
478 measure of upper-tail fidelity. The 0.99-quantile error is

479
$$e_{0.99} = \left| \frac{\widehat{q}_{0.99} - q_{0.99}}{q_{0.99}} \right|. \quad (25)$$

480 Because each fitted distribution is reused across tariff cells within a history, the $q_{0.99}$ and mean errors
481 are history-level quantities. Stop-loss error is cell-specific because its oracle evaluation threshold varies
482 with the tariff cell; for method-level fidelity summaries, the nine cell-specific errors are averaged with
483 equal weight within history. The 10,000 histories are the independent Monte Carlo units; across five meth-
484 ods they yield 50,000 history–method predictions, with methods paired within history. Pairwise fidelity
485 comparisons preserve this within-history pairing.

486 Decision performance is evaluated separately through tariff-mode correctness and full-action regret as
487 defined in Section 3.2. In cells with a unique oracle mode, correctness records whether $\widehat{m} = m_F^*$. Full-action
488 regret is reported both in CNY/month and relative to the cell-specific oracle value, and additionally cap-
489 tures contract-level error when contract demand is selected. For regional comparisons, cells are grouped
490 by oracle tariff mode and the three cells in each region are averaged with equal weight within history and
491 method. This gives 10,000 paired history-level differences for each regional method contrast. Contrasts
492 are defined as the first method minus the second, so negative values indicate lower regret for the first
493 method. Figure 2(b–d) shows TAIL–EVENT–EMP, TAIL–KDE, and GEV–KDE. Table S12 reports all ten
494 pairwise method contrasts within each of the three oracle regions.

495 The formal normalized regret is

496
$$\widetilde{R} = \frac{R_F(\widehat{a})}{V_F^*}.$$

507 The regional estimand is the equal-cell average of the paired full-action-regret differences over the prespec-
 508 ified cells within each oracle region. Formal inference targets its mean on the absolute CNY/month scale.
 509 Uncertainty is summarized by paired Student t 95% intervals across the 10,000 histories ($df = 9,999$).

500 Tariff-mode-error diagnostics are evaluated at the individual tariff-decision-row level. The nine cells,
 501 five methods, and 10,000 histories produce

$$502 \quad 9 \times 5 \times 10,000 = 450,000$$

503 decision rows. Tariff-mode error is the positive class, and larger diagnostic values are treated as stronger ev-
 504 idence of error. The absolute relative mean error, measuring error in the distributional functional entering
 505 the actual-demand cost, is

$$506 \quad e_\mu = \left| \frac{\mu_{\hat{F}} - \mu_F}{\mu_F} \right|. \quad (26)$$

507 Let $D_{F,\text{con}}^*$ denote the oracle contract level returned by the within-contract-mode optimization rule,
 508 irrespective of which tariff mode is optimal overall, and define

$$509 \quad t_{F,\text{con}}^* = \alpha D_{F,\text{con}}^*.$$

510 At this oracle contract threshold, the absolute relative stop-loss error is

$$511 \quad e_H = \left| \frac{H_{\hat{F}}(t_{F,\text{con}}^*) - H_F(t_{F,\text{con}}^*)}{H_F(t_{F,\text{con}}^*)} \right|. \quad (27)$$

512 The diagnostic e_H evaluates error at one oracle threshold per cell. The bound in Equation (11) uses the
 513 supremum of the absolute stop-loss error over all feasible thresholds. Thus e_μ and e_H evaluate errors in
 514 tariff-relevant distributional functionals, whereas $e_{0.99}$ serves as a complementary upper-tail comparator.
 515 The fourth diagnostic is the optimized-mode-value error relative to oracle mode separation,

$$516 \quad d_{MV} = \frac{2 \max_{m \in \mathcal{M}} |\hat{V}(m) - V_F(m)|}{\gamma_F}.$$

517 All controlled cells have positive oracle mode gaps, so no zero-denominator adjustment is required. The
 518 quantity d_{MV} evaluates the errors of the three internally optimized mode values relative to oracle mode
 519 separation. Its numerator concerns optimized mode-value errors, whereas $\varepsilon(F, \hat{F})$ is defined over the
 520 primitive-action space; the two quantities therefore have distinct estimands. Its primary assessment is the
 521 structural certificate $d_{MV} < 1 \Rightarrow \hat{m} = m_F^*$, checked row by row.

522 To assess association with economic regret, regret and the cell-specific stop-loss error are averaged
 523 over the nine cells within each history and method; the history-level $q_{0.99}$ and mean errors are carried
 524 unchanged to the same analysis unit. Spearman correlations are computed separately for each method.
 525 Within each method, the mean- and stop-loss-error correlations are compared with the $q_{0.99}$ -error corre-
 526 lation, giving ten contrasts across the five methods. The estimand is the difference between the signed
 527 correlation coefficients. Simultaneous familywise 95% intervals and single-step adjusted p -values for this
 528 ten-contrast family are obtained from a centered, studentized max- $|T|$ procedure using 2,000 paired boot-
 529 strap resamples of the 10,000 histories. The same resampled history identifiers are used across all methods
 530 and contrasts, with ranks and correlations recomputed in each resample.

531 Independent numerical checks of the contract optimizer, including boundary and flat-optimum configu-
532 rations, are reported in Appendix S1.15.

533 4.3 | Robustness designs

534 Additional designs examine tariff geometry, fitted-support sensitivity, high- p^* tariff targets, and alternative
535 data-generating mechanisms.

536 A paired sensitivity analysis additionally uses the first 1,000 primary histories to compare the fitted-
537 support specification used in the primary analysis with training-derived upper bounds $c \max_{1 \leq m \leq 24} M_m$, for
538 $c \in \{10, 20, 50\}$; details and cell-level results are reported in the Supporting Information.

539 The crossed grid defined in Section 4.1 supplies the geometry sensitivity analysis. Nearest-gap com-
540 parisons rank cross-region cell pairs by relative oracle-gap distance; their descriptive construction and
541 observed distances are reported in the Supporting Information.

542 A tariff-parameter stress keeps the baseline data-generating distribution and the fitted laws from all
543 10,000 primary histories fixed while shifting the contract-mode target quantile into the upper tail. With
544 $\alpha = 2.20$, κ is chosen so that

545
$$p^* = 1 - \frac{1}{\kappa\alpha} \in \{0.90, 0.95, 0.99\},$$

546 and ten utilization cells at each level span actual-demand, contract-demand, and capacity oracle regions.
547 When $p^* = 0.99$, an interior contract optimum targets $q_{0.99}$ directly, whereas boundary optima remain
548 determined by the feasible contract interval. Complete cell-level and same-information paired results are
549 reported in Figure S1 and Table S1.

550 A joint count–severity misspecification experiment replaces the baseline Poisson–GPD tail mechanism
551 by negative-binomial tail counts and Weibull excesses while retaining the 24-month training design, five
552 primary predictive constructions, and nine tariff cells. The tail-count distribution has mean 30 and size
553 parameter 1, and excesses above the 80,000 kW generating threshold follow a Weibull distribution with
554 shape 1.5 and scale 30,000 kW. The bulk component contains 200 events per month with demand gener-
555 ated between 45,000 and 80,000 kW using a Beta(8, 2) construction. The experiment uses 10,000 histories,
556 giving 450,000 method–history–cell decisions. It measures the combined effect of count and excess-law
557 misspecification on the downstream tariff decision.

558 A separate physical-pipeline stress examines high-amplitude, high-frequency, and clustered-event mech-
559 anisms at $\alpha = 1.20$ using 12- and 24-month histories. Its distinct predictive constructions, oracle tie
560 treatment, and cell-level results are reported in Figure S6 and Table S3.

561 Other supplementary experiments examine parameter recovery, coverage of threshold-fixed boot-
562 strap intervals, and physical clipping; their distinct settings and results are reported in the Supporting
563 Information.

564 4.4 | External manufacturing-load analysis

565 External manufacturing-load evidence was obtained from the public South Korean manufacturing-load
566 dataset of Lee, Baek and Kim¹³ (Figshare Version 9 release; doi:10.6084/m9.figshare.14822256.v9)¹⁴,
567 comprising one-minute electricity-demand records in kW for 10 manufacturing facilities from March to
568 September 2019. All 10 facilities are retained for descriptive comparison of upper-tail behaviour, with

load normalized by the corresponding factory mean before empirical survival probabilities are calculated. Formal forecasting uses fixed, non-overlapping 15-minute averages. For the primary external analysis, documented demand-response event intervals and the subsequent 60-minute rebound window are excluded; corresponding analyses retaining all observations are reported in the Supporting Information. A reconstructed window is retained only when all 15 one-minute observations are available, with no imputation in the primary series, and a factory-month is eligible when at least 95% of reconstructed windows are valid. Four expanding-window origins are evaluated: March–May to June, March–June to July, March–July to August, and March–August to September. An origin is eligible only when all required training and test months satisfy the primary completeness criterion. Eight factories satisfy the complete formal evaluation rule; Metal 2 and Steel 1 remain in the descriptive analysis but are not assigned formal predictive scores.

For each factory and origin, let $M_{o,m}$ denote the largest reconstructed 15-minute window in training month m . The normalization scale is

$$s_o = \frac{1}{|T_o|} \sum_{m \in T_o} M_{o,m},$$

the mean of the training monthly maxima, and the same scale is applied to the training windows and the test month. The external comparison evaluates seven predictive constructions: K-BLOCK, K-EMP, K-GEV, K-GTAIL, K-KDE, K-POINT, and K-UGPD. K-GTAIL candidate specifications are selected by leave-one-training-month-out validation, with normalization recomputed within each fold; the selected construction is then refitted to all training months before the test-month forecast. The external scoring representations use 50,000 predictive draws where applicable. Method-specific fitting, support checks, and fallback rules are detailed in Appendix S1.13.

Predictive performance is evaluated using CRPS, qwCRPS_{0.90}, and pinball losses at $q_{0.95}$ and $q_{0.99}$. The fixed upper-quantile grid used for qwCRPS_{0.90} is specified in Appendix S1.13. Empirical exceedance frequencies at each method’s predictive quantiles provide supplementary calibration summaries. The continuous ranked probability score is

$$\text{CRPS}(F, y) = \mathbb{E}|X - y| - \frac{1}{2} \mathbb{E}|X - X'|.$$

For each score endpoint s and baseline method b , define

$$G_{b,s} = L_{\text{K-GTAIL},s} - L_{b,s}.$$

Negative values therefore favour K-GTAIL because lower predictive scores indicate better performance. The four origin-specific differences are first averaged within each of the eight eligible factories. Uncertainty in the overall mean difference is summarized by a paired Student t 95% interval across the eight factory-level differences ($df = 7$).

5 | RESULTS

5.1 | Tariff geometry creates heterogeneous decision sensitivity

Oracle tariff modes and their separation changed substantially across the prespecified tariff geometries. Figure 1(a,b) shows this structure for $\alpha = 1.20$, where the optimized oracle mode moves across actual-demand, contract-demand, and capacity charging as utilization increases. Across the full nine-cell primary

design, the relative oracle gap,

$$100\gamma_F/V_F^*,$$

ranged from 0.093% to 10.732%, spanning more than two orders of magnitude. The smallest separations occurred near tariff-mode boundaries: at $\alpha = 1.20$, the actual-demand cell at $u = 0.450$ had a relative gap of 0.093%, the contract-demand cell at $u = 0.455$ had a gap of 0.223%, and the capacity cell at $u = 0.635$ had a gap of 0.450%. By comparison, the actual-demand cell at $u = 0.400$ and the capacity cell at $u = 0.700$ were more clearly separated, with relative gaps of 5.971% and 10.732%, respectively. The remaining primary-cell oracle quantities, including the two $\alpha = 1.05$ configurations, are reported in Table S13.

The same variation in tariff geometry was accompanied by materially different full-action regret profiles. Figure 1(c) reports mean regret for all five predictive constructions across all nine primary cells, including seven at $\alpha = 1.20$ and two at $\alpha = 1.05$, using 10,000 histories for each method–cell combination. Mean regret and the between-method differences varied substantially across cells as the design moved through actual-demand, contract-demand, and capacity regions. These differences were visible both around the closely spaced contract–capacity transition and at more clearly separated configurations. These cell-level differences motivate evaluating distributional fidelity together with the downstream economic loss generated by the resulting tariff action.

5.2 | Economic rankings vary with tariff geometry

Statistical fidelity did not produce a single economic ordering. Under the controlled data-generating process, TAIL had the smallest median absolute relative $q_{0.99}$ error, 10.09%, compared with 14.30% for KDE (Table 2; Figure 2(a)). The median stop-loss errors, evaluated at the cell-specific oracle contract thresholds and averaged over the nine cells within history, were 28.49% for TAIL, 31.00% for GEV, and 33.55% for KDE. The association analysis showed a corresponding functional pattern. For TAIL, Spearman correlations of regret with absolute relative mean, stop-loss, and $q_{0.99}$ errors were 0.753, 0.683, and 0.534. The mean-minus- $q_{0.99}$ and stop-loss-minus- $q_{0.99}$ correlation differences were 0.220 and 0.149, with simultaneous familywise 95% intervals of [0.203, 0.237] and [0.139, 0.160]. All ten simultaneous intervals comparing mean or stop-loss error with $q_{0.99}$ error were positive across the five predictive methods (Table S7).

Within the monthly-maxima information set, GEV and KDE reversed their economic ordering across oracle regions. The GEV-minus-KDE mean-regret contrast was +0.822 thousand CNY/month in the actual-demand region (95% interval [0.677, 0.967]), −3.887 in the contract-demand region ([−4.143, −3.632]), and +5.470 in the capacity region ([4.874, 6.067]). Within the event-level information set, TAIL-minus-EVENT-EMP contrasts were −0.606, −1.680, and −2.671 thousand CNY/month in the same three regions, with all three 95% intervals excluding zero. Complete regional pairwise contrasts are reported in Table S12.

The complete-procedure TAIL–KDE comparison, which also differs in information resolution, showed the same dependence on tariff geometry. Its regional mean contrasts were −0.659, −7.379, and +4.129 thousand CNY/month in the actual-demand, contract-demand, and capacity regions. The capacity-region average combined markedly different cell-level effects: +1.004, −1.561, and +12.945 thousand CNY/month at $(\alpha, u) = (1.05, 0.630)$, $(1.20, 0.635)$, and $(1.20, 0.700)$. At the last cell, TAIL and KDE selected

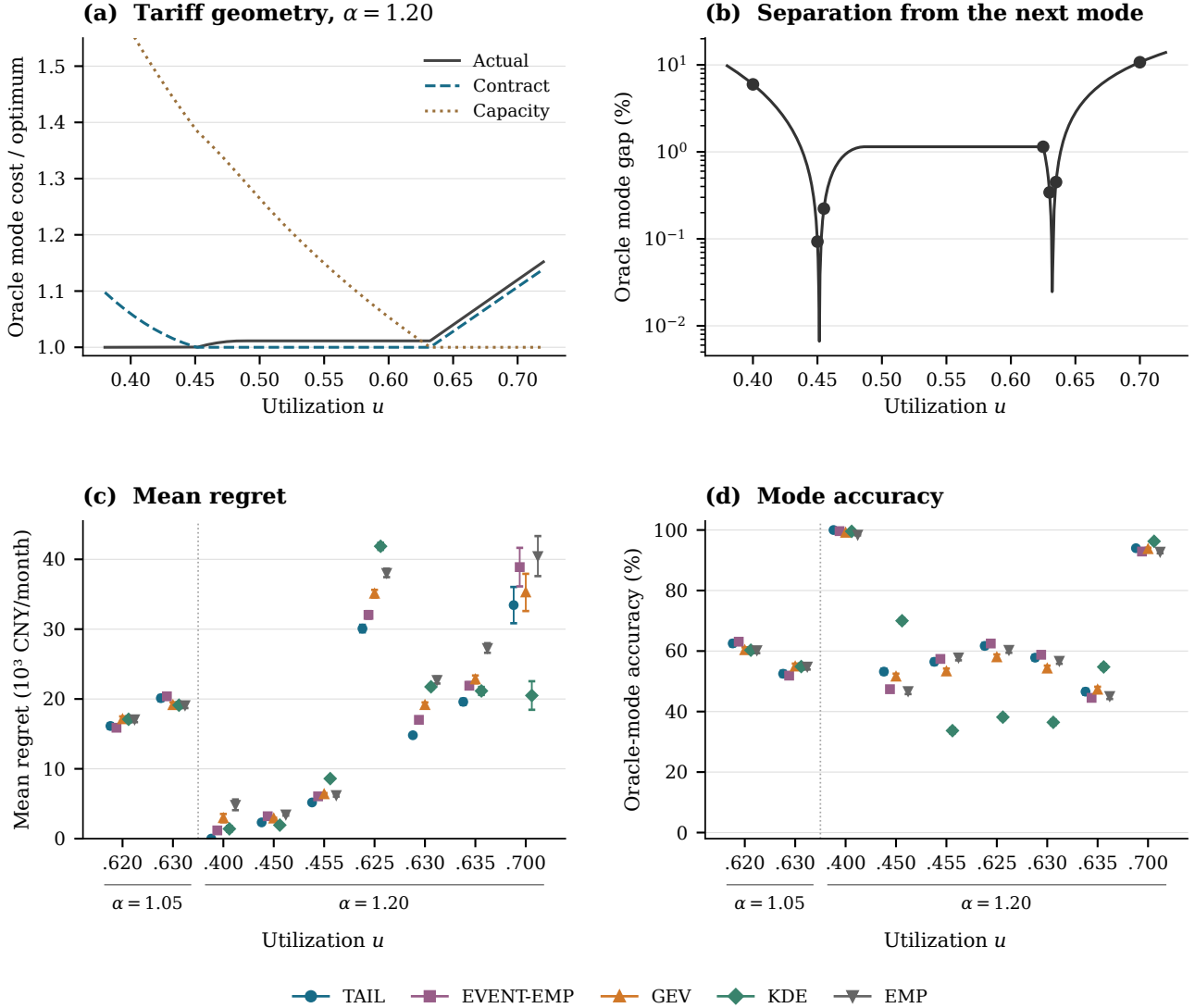


FIGURE 1 Analytic tariff geometry and primary decision performance. (a) Oracle optimized mode costs divided by the best oracle value and (b) the relative gap between the two lowest oracle mode values, at $\alpha = 1.20$; dots in (b) identify the seven primary cells at this α . The seven displayed cells form the $\alpha = 1.20$ subset of the nine-cell primary tariff design summarized in Table S13. (c) Mean full-action regret and pointwise t 95% intervals and (d) oracle-mode accuracy with Wilson 95% intervals for five methods at nine prespecified cells. Each method–cell summary uses 10,000 independent 24-month histories; fitted laws and histories are shared across cells. TAIL and EVENT-EMP use event histories; GEV, KDE and EMP use monthly maxima. The lower panels use equally spaced cell positions, labelled by utilization and grouped by α , rather than a continuous utilization axis.

645 a non-oracle mode in 600 and 373 of the 10,000 histories. Because the oracle action there is the singleton
 646 capacity mode, every mode-selection error incurs a discrete mode loss; selections of contract demand can
 647 additionally retain within-contract loss. Table 3 reports the corresponding method-specific regional regret
 648 and oracle-mode accuracy.

649 The geometry dependence remained visible beyond the nine-cell primary design. Figure 2(e–h)
 650 evaluates four values of α and 19 utilization levels per value, giving 76 tariff cells and 380,000 method–
 651 history–cell evaluations. Both the TAIL–KDE and GEV–KDE cellwise mean-regret contrasts varied with

utilization and oracle mode. Oracle contract demand occurred in nine crossed-grid cells, all at $\alpha = 1.20$; the remaining cells were assigned to actual demand or capacity. These profiles are descriptive at the cell level, with history retained as the sampling unit for inferential comparisons. Complete five-method normalized-regret and oracle-mode-accuracy profiles are reported in Figure S8.

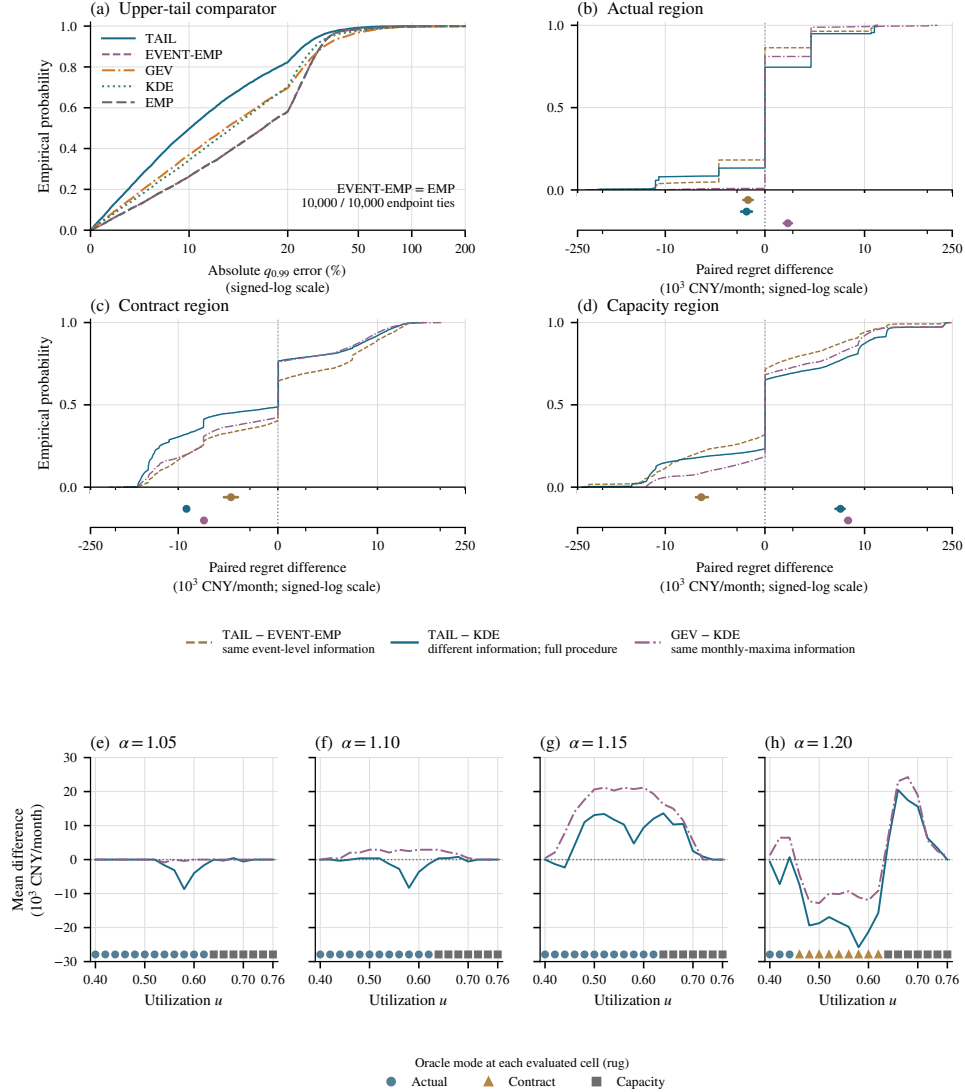


FIGURE 2 Statistical fidelity and geometry-dependent regret. (a) Empirical CDFs of absolute relative $q_{0.99}$ error over 10,000 histories per method; EVENT-EMP and EMP coincide. The scale is linear to 20% and logarithmic above. (b)–(d) Paired full-action regret differences for the actual-demand, contract-demand, and capacity regions, respectively, after averaging the three cells within each of 10,000 histories. Contrasts are TAIL–EVENT-EMP, TAIL–KDE, and GEV–KDE; negative values favour the first method. ECDFs summarize the paired differences, with mean contrasts and paired Student t 95% intervals ($df = 9,999$) shown as formal inferential summaries. Axes are linear within ± 2 thousand CNY/month and logarithmic outside. (e)–(h) Descriptive mean-regret contrasts over the 76-cell crossed grid ($\alpha = 1.05, 1.10, 1.15, 1.20$; 19 utilization levels and 1,000 histories per method–cell). Solid and dash-dotted lines denote TAIL–KDE and GEV–KDE; rugs show oracle modes. Complete five-method crossed-grid profiles are in Figure S8, and primary regional paired contrasts are reported in Table S12.

5.3 | Tariff contrasts sharpen demand-scale stability

The competitor-specific mode-value condition held for 19.52% of the 450,000 primary decisions, compared with 10.208% for the scalar $d_{MV} < 1$ condition (Figure 3(a,b)). Neither set contained an observed tariff-mode error. The difference reflects the information retained by the two conditions: the competitor-specific condition keeps each oracle gap and optimized mode-value error separate, whereas d_{MV} reduces them to a single conservative ratio.

The demand-scale analysis shows a parallel hierarchy. The contrast-aware radius was never smaller than the mode-wise pairwise radius and was strictly larger in four of the nine primary tariff cells (Table S13). At $(\alpha, u) = (1.20, 0.400)$, $(1.20, 0.450)$, and $(1.20, 0.455)$ it was three times larger; at $(1.20, 0.625)$ it was 1.5 times larger because the binding competitor changed from actual demand to capacity. Using the same numerical W_1 upper quantities and solver allowance, the contrast-Wasserstein check held for 57,501 decisions (12.778%), compared with 26,393 (5.865%) for the mode-wise pairwise-Wasserstein check and 5,329 (1.184%) for the uniform-Wasserstein check. No observed mode-selection violation occurred among rows satisfying any of these numerical conditions (Table S14). The numerical W_1 quantities and solver allowances are floating-point constructions rather than directed-rounding interval bounds.

The exact regret decomposition separates preservation of the tariff mode from preservation of the complete action. Actual-demand and capacity modes have no within-mode degree of freedom, so selecting either oracle-optimal mode gives zero within-mode excess. Contract demand retains the continuous choice of contract level. Figure 3(c,d) therefore restricts residual loss to decisions for which contract demand is both oracle optimal and selected. Positive within-contract regret can remain even when the tariff mode is preserved.

5.4 | Stress designs reveal mechanism-dependent rankings

The joint negative-binomial–Weibull challenge preserved the region-dependent economic comparisons under a different count and severity mechanism (Figure S7; Tables S4–S5). For the same-information event-level comparison, TAIL-minus-EVENT-EMP mean-regret contrasts were -28.97 , -11.71 , and $+7.55$ thousand CNY/month in the actual-demand, contract-demand, and capacity regions. For the monthly-maxima comparison, GEV-minus-KDE contrasts were $+5.06$, -34.32 , and $+1.98$ thousand CNY/month across the same regions. All six 95% intervals excluded zero. The two information-matched comparisons therefore produced different economic orderings as the oracle tariff region changed.

The physical-pipeline experiment produced a different ordering under high-amplitude, high-frequency, and clustered-event mechanisms (Figure S6; Table S3). With 24 months of training, mean regret for KDE-MC-JOINT was 0.842, 0.948, and 1.020 thousand CNY/month across the three scenarios, compared with 4.451, 4.311, and 3.782 for TAIL-JOINT. Tail-oriented construction therefore did not translate into the lowest downstream regret under these alternative physical mechanisms.

The high- p^* analysis varied the tariff target while retaining the same baseline histories and fitted laws (Figure S1; Table S1). The GEV-minus-KDE contract-region contrast changed direction between $p^* = 0.90$ and 0.95 , and its pointwise 95% interval included zero at $p^* = 0.99$; TAIL-minus-EVENT-EMP remained negative at all three levels. At $p^* = 0.99$, two of the four contract-region cells had interior oracle

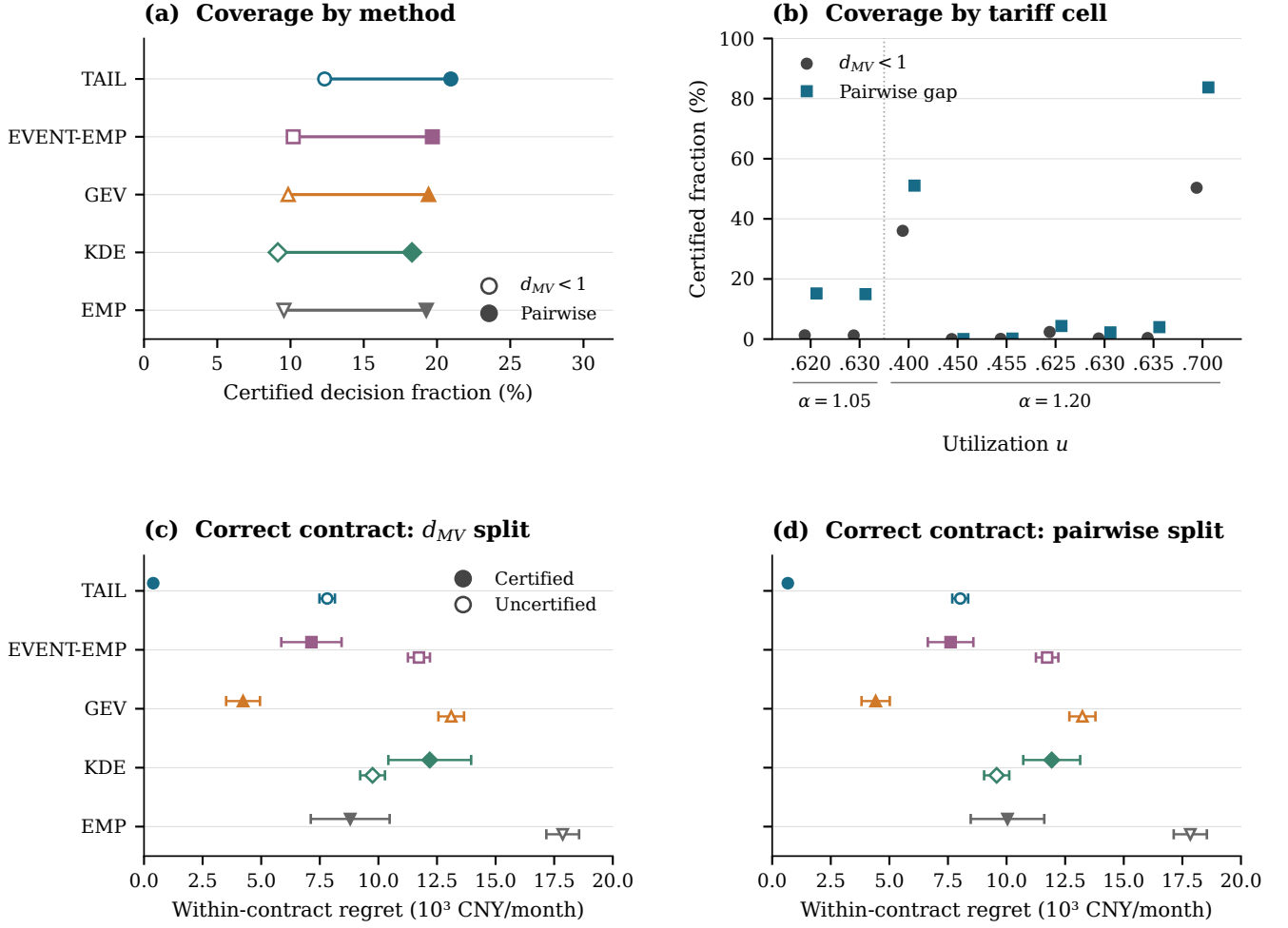


FIGURE 3 Mode-value stability and residual within-contract regret. The primary analysis contains 450,000 method–history–cell decisions from 10,000 histories, five predictive methods, and nine tariff cells. **(a)** Decision fractions satisfying the scalar $d_{MV} < 1$ and competitor-specific mode-value sufficient conditions, by predictive method. **(b)** Corresponding fractions across the nine primary tariff cells. **(c)** Method-specific within-contract regret among decisions for which contract demand is both oracle optimal and selected, split by the $d_{MV} < 1$ condition. **(d)** The same correct-contract subset split by the competitor-specific mode-value condition. Panels (c,d) omit mode-correct actual-demand and capacity decisions because their within-mode excess is identically zero. Within-contract regret is shown in 10^3 CNY/month.

694 optima and two had lower-bound optima. The $q_{0.99}$ target is therefore exact for the interior cells, whereas
 695 region-level full-action regret continues to reflect the complete tariff decision.

696 Fitted-support sensitivity was limited over the first 1,000 paired primary histories. Alternative support
 697 specifications changed no more than 0.20% of TAIL mode decisions or 0.40% of GEV mode decisions
 698 in any cell; the largest absolute cellwise paired mean-regret changes were 0.062 and 0.726 thousand
 699 CNY/month (Figure S3(d–f); Table S11). Parameter-recovery, threshold-fixed bootstrap coverage, and
 700 physical-clipping analyses are reported in Figures S4–S5 and Table S10.

5.5 | External rankings vary across predictive and tariff targets

The manufacturing-load data produced different point-estimate rankings across predictive targets (Figure 4(a–d)). K-BLOCK and K-KDE had lower mean CRPS than K-GTAIL (0.025390 and 0.026363 versus 0.031017). For the upper tail, K-GTAIL had the lowest of these three mean $\text{qwCRPS}_{0.90}$ values (0.001204 versus 0.001290 for K-KDE and 0.001713 for K-BLOCK) and the lowest $q_{0.99}$ pinball loss (0.003212 versus 0.003262 and 0.007283).

Factory-level uncertainty was large relative to these point-estimate differences. The K-GTAIL-minus-K-BLOCK CRPS contrast was 0.005627, with a paired t 95% interval of $[-0.009534, 0.020789]$ across the eight factories ($df = 7$). All 16 pointwise intervals comparing K-GTAIL with the four baselines across the four predictive losses included zero (Table S9). The external predictive results therefore show target-dependent point rankings together with substantial between-factory uncertainty.

Figure 4(e,f) maps the same forecasts to hypothetical tariff actions. The panels report normalized next-month charge contrasts for K-GTAIL versus K-BLOCK and K-GEV versus K-KDE across the nine tariff cells, using the observed next-month maximum to evaluate the selected action. The contrasts vary across tariff cells and their factory-level intervals are wide relative to the cellwise means, providing a decision-linked complement to the predictive-score comparison.

Fit-status summaries in Table S8 show that the prespecified fallback branches remained active in the rolling procedures. The external scores and charge contrasts therefore represent the complete predictive procedures, rather than only origins at which the primary tail fits were available.

The K-GEV minimum-sample rule assigned the empirical fallback to the three-, four-, and five-month training origins for all eight factories, accounting for all 24 fallback predictions. All eight six-month K-GEV fits passed. K-GTAIL used its primary fit in 15 factory–origin units, an exponential-tail fallback in 13, and an empirical fallback in four. All three fitting branches contributed to the reported rolling-origin scores.

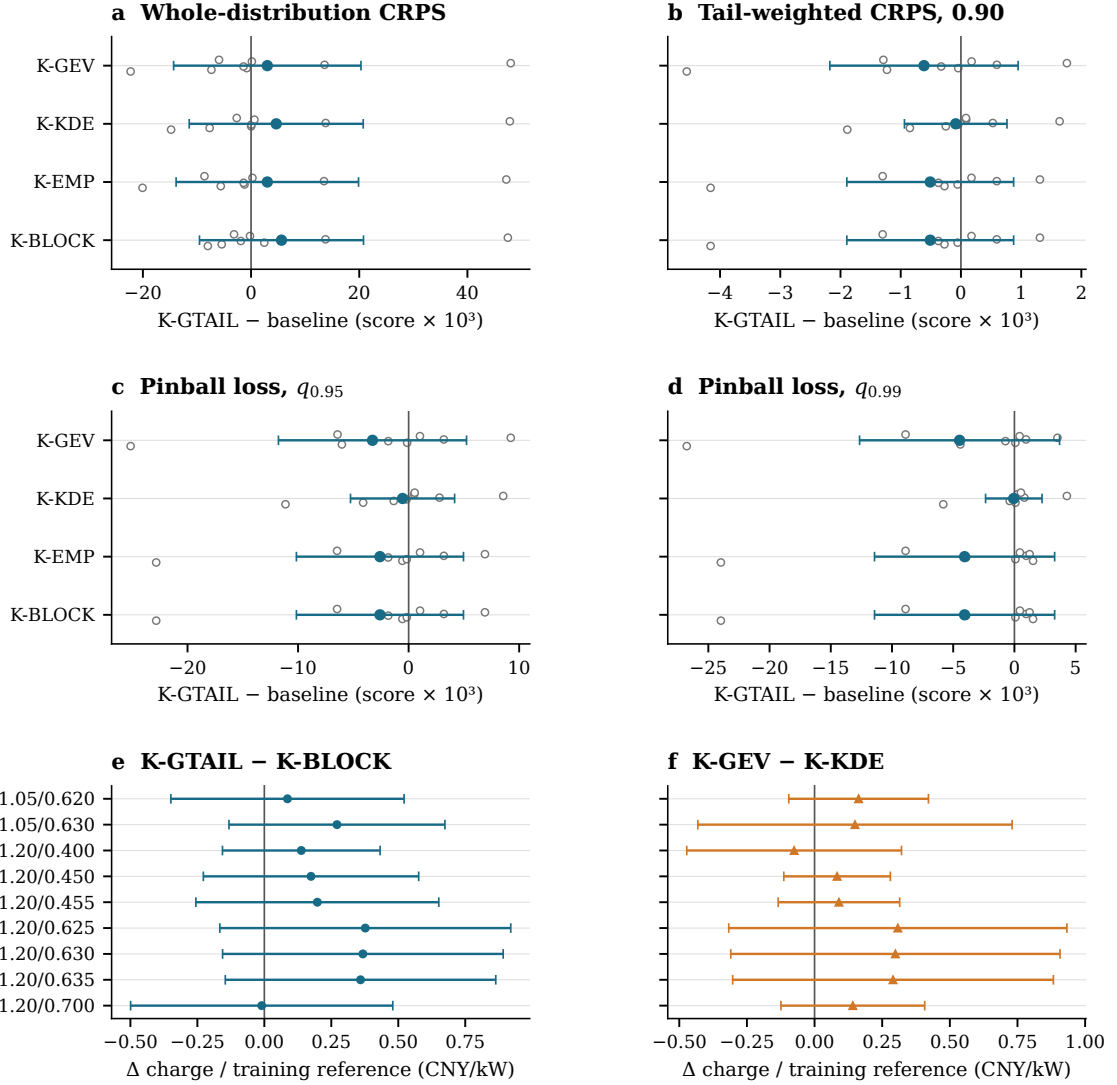


FIGURE 4 Factory-level predictive-score and hypothetical-charge comparisons. The external analysis uses eight factories with four rolling origins per factory. (a–d) K-GTAIL-minus-baseline differences in CRPS, 0.90-tail-weighted CRPS, and pinball loss at the 0.95 and 0.99 quantiles. Open dots are factory-specific differences after averaging the four origins; filled dots and bars are the mean and paired Student t 95% interval across the eight factories ($df = 7$). Lower predictive scores are better. (e,f) Differences in hypothetical next-month demand charges, divided by the training reference demand, across the nine prespecified tariff cells. Each charge is obtained by evaluating the selected action at the observed next-month maximum. The same within-factory averaging and paired t construction are used. Negative differences favour the first named method. Oracle regret is defined for the controlled experiment; panels (e,f) report charge contrasts evaluated on the external load records.

725 6 | DISCUSSION

726 Tariff geometry, rather than a global forecast ranking, organizes the economic comparisons in the con-
 727 trolled study. Within the shared monthly-maxima information set, the GEV–KDE contrast favours KDE
 728 in the actual-demand region, GEV in the contract-demand region, and KDE again in the capacity region.
 729 The complete-procedure TAIL–KDE comparison displays the same region dependence, with substantial

cell-level heterogeneity inside the capacity region. This pattern is consistent with recent decision-focused and application-driven forecasting studies in which statistical forecast quality and downstream economic value need not induce the same ranking^{8–11}. In the present tariff problem, the mechanism is explicit: tariff geometry changes both the distributional functional entering expected cost and the separation between competing optimized mode values.

The contract mode makes this functional dependence particularly clear. Its tariff coefficients determine the probability level $p^* = 1 - 1/(\kappa\alpha)$ and the feasible threshold interval, so the optimized contract problem has a constrained Expected-Shortfall representation. A generalized p^* -quantile locates an optimal threshold, while the associated upper-tail average determines the optimized contract value whenever the quantile-optimal set is feasible. The primary interior targets correspond to probabilities near 0.524 and 0.583, rather than 0.99, which explains why absolute mean and stop-loss errors are more closely associated with regret than $q_{0.99}$ error in the baseline design. The high- p^* analysis moves the tariff target into the upper tail and shows that the same quantile can have different decision implications when feasibility constraints change the contract optimum. Forecast value therefore depends on the functional selected by the decision problem, a principle also emphasized in recent operational evaluations of probabilistic forecasts^{7,8}.

Mode separation determines how much distributional perturbation the discrete tariff choice can absorb. Bounding the optimized mode values separately gives a valid Wasserstein condition, but direct comparison of actual demand with contract demand retains cancellation because both costs respond to the same predictive-law perturbation. The resulting contrast-aware radius is no smaller than the mode-wise pairwise radius and is strictly larger in four of the nine primary cells, by as much as a factor of three. The corresponding numerical sufficient condition is satisfied by 12.778% of primary decisions, compared with 5.865% under the separate mode-value construction, with no observed mode-selection error in either passing set. The exact regret decomposition adds a different distinction: preserving the tariff mode does not necessarily preserve the full action, because contract demand retains a continuous contract-level decision after the mode itself has been selected.

The stress designs show how these mechanisms interact with the predictive construction and the data-generating process. Under the negative-binomial–Weibull challenge, both information-matched comparisons change economic ordering across oracle regions. Under the physical-pipeline mechanisms, KDE-MC-JOINT has lower mean regret than TAIL-JOINT in all three 24-month scenarios. The high- p^* and fitted-support analyses separate tariff-target and support sensitivity from these distributional changes. The manufacturing-load analysis adds observed industrial demand: whole-distribution and upper-tail scores produce different point rankings, while factory-level uncertainty remains substantial and the hypothetical-charge contrasts vary across tariff cells. Across these settings, the downstream objective does not support a single predictive construction uniformly.

The analytic Poisson–GPD design makes the complete propagation from distributional estimation to tariff action observable against an exact oracle. The alternative count–severity and physical-pipeline designs show how the economic comparisons change under different mechanisms. The external analysis adds rolling forecasts of observed industrial load and evaluates one-month hypothetical charges under the same tariff family. Longer records linked to actual contract choices and billing outcomes would extend the present decision-linked evaluation to realized industrial costs, while covariate-dependent peak-demand models would allow production state and other operating conditions to enter the predictive law directly¹⁵.

722 7 | CONCLUSIONS

723 This study develops a tariff-specific account of how distributional estimation error propagates into
724 industrial electricity decisions. Contract-demand optimization has a constrained Expected-Shortfall repre-
725 sentation at a tariff-implied probability level, full-action regret separates mode-selection loss from residual
726 within-contract loss, and direct tariff contrasts yield competitor-specific Wasserstein stability margins that
727 retain shared distributional perturbations.

728 The paired experiments show that economic model rankings can change with tariff geometry even
729 when the compared procedures use the same information source. Alternative count-severity and physical-
730 pipeline mechanisms also change relative performance, while the manufacturing-load analysis produces
731 target-dependent predictive rankings and uncertain hypothetical-charge contrasts across factories. Taken
732 together, the results support evaluating peak-demand forecasts through the tariff-specific functionals,
733 competing cost margins, and actions that determine their economic value.

734 SUPPORTING INFORMATION

735 Figures S1–S8 present, respectively, the high- p^* tariff stress, functional-error and certificate diag-
736 nostics, numerical distribution-to-decision bounds with fitting-support sensitivity, parameter recovery,
737 physical-clipping sensitivity, physical-pipeline stress, the joint negative-binomial–Weibull challenge, and
738 crossed tariff-grid profiles. Tables S1–S14 provide the corresponding high- p^* comparisons, certificate-
739 conditioned contract losses, physical-pipeline results, negative-binomial–Weibull results, Wasserstein
740 numerical quantities, functional-error contrasts, external predictive summaries, threshold-fixed bootstrap
741 coverage, fitting-support sensitivity, primary regional paired contrasts, analytic demand-scale geometry,
742 and numerical sufficient-condition audit.

743 DATA AND CODE AVAILABILITY

744 The controlled analyses in this study are based on synthetic data generated from the stochastic mechanisms
745 described in Sections 4.1 and 4.3. The South Korean manufacturing-load data used for the external analysis
746 are publicly available from Lee, Baek and Kim¹³ and the accompanying Figshare release¹⁴.

747 The simulation and model-fitting code, tariff calculations, external preprocessing, experiment config-
748 urations, random seeds, and result data are available at [https://github.com/riddlehuang05/extre](https://github.com/riddlehuang05/extreme-demand-tariff-analysis)
749 me-demand-tariff-analysis. The repository documents the supplied workflows and dependencies in
800 docs/REPRODUCING.md and docs/RELEASE_SCOPE.md. Its data/ directory contains indexed
801 source-data files, including functional-error analyses and solver comparisons. The accompanying source
802 manifest, derivation map, and figure/table manifests document source records and panel mappings for the
803 supplementary figures and tables; the same source files accompany the article as a source-data supplement.
804 External source data are obtained from the cited Figshare release under its access terms.

805 **CONFLICT OF INTEREST STATEMENT**

806 The authors declare no conflicts of interest.

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811 **AI-ASSISTED PREPARATION**

812 OpenAI Codex was used to assist with code development and checking, analysis implementation,
813 manuscript revision, language editing, and preparation of submission materials. The authors take
814 responsibility for the content and interpretation of the work.

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TABLE 1 Tariff modes, within-mode decisions, and distributional functionals.

Tariff mode	Within-mode decision	Expected monthly cost	Distributional functional
Capacity	None	$r_c S$	None
Actual demand	None	$r_d \mu_F$	Mean μ_F
Contract demand	$D \in [\delta S, S]$	$r_d \{D + \kappa H_F(\alpha D)\}$	Stop-loss $H_F(\alpha D)$

Notes. The demand quantities M , D , and S are defined on a common basis. For contract demand, varying D over $[\delta S, S]$ requires the stop-loss transform over the corresponding feasible threshold set $\{\alpha D : D \in [\delta S, S]\}$. After optimization, the contract-demand objective has the constrained Expected-Shortfall representation in Equation (5); the stop-loss curve is the primitive functional from which this optimized value is obtained.

TABLE 2 Distributional and tariff-functional fidelity in the primary experiment.

Method	$q_{0.99}$ error (%)		Mean error (%)		Stop-loss error (%)	
	Median	Mean	Median	Mean	Median	Mean
TAIL	10.09	12.22	2.82	3.34	28.49	34.02
EVENT-EMP	17.52	18.17	2.79	3.36	30.95	37.35
GEV	13.73	16.63	2.98	3.55	31.00	38.03
KDE	14.30	16.17	2.98	3.56	33.55	44.95
EMP	17.52	18.17	2.98	3.55	31.55	38.24

Notes. Entries summarize absolute relative errors over 10,000 independent 24-month histories. For stop-loss error, the nine cell-specific oracle-threshold errors are first averaged within history; median and mean are then taken across histories. Quantile and mean estimates are shared across cells. TAIL and EVENT-EMP use event histories; GEV, KDE and EMP use monthly maxima. EVENT-EMP and EMP have identical left $q_{0.99}$ estimates, at the observed upper support endpoint.

TABLE 3 Regional decision loss across tariff modes.

Region	Method	Mean regret (thousand CNY/month)	Mean normalized regret (%)	Oracle-mode accuracy (%)
Actual-demand	TAIL	6.15	0.115	71.9
Actual-demand	EVENT-EMP	6.76	0.127	70.0
Actual-demand	GEV	7.63	0.143	70.3
Actual-demand	KDE	6.81	0.128	76.6
Actual-demand	EMP	8.47	0.159	68.5
Contract-demand	TAIL	16.69	0.316	58.7
Contract-demand	EVENT-EMP	18.37	0.348	59.6
Contract-demand	GEV	20.18	0.382	55.1
Contract-demand	KDE	24.07	0.456	36.1
Contract-demand	EMP	22.34	0.423	58.4
Capacity	TAIL	24.38	0.485	64.4
Capacity	EVENT-EMP	27.05	0.539	63.1
Capacity	GEV	25.73	0.512	65.2
Capacity	KDE	20.25	0.398	68.6
Capacity	EMP	28.96	0.577	64.3

Notes. Each oracle region contains three prespecified primary tariff cells. Within each of the 10,000 histories and each predictive method, cell-level quantities are first averaged over the three cells in the region and are then averaged across histories. Mean regret is full-action regret $R_F(\hat{a})$, reported in 10^3 CNY/month. Mean normalized regret is $100R_F(\hat{a})/V_F^*$. Oracle-mode accuracy is the percentage of corresponding cell-level decisions satisfying $\hat{m} = m_F^*$. Formal paired regional mean differences and their paired Student t 95% intervals are shown in Figure 2(b–d) and reported in Table S12.